

LSI-lite: A Composition Analysis Tool (with Color Telemetry)

Profiled Composition Structural Exploration (Δx , r_v , ρ_r)

What It Does

LSI-lite measures how an image **behaves under compositional structure** using three primitives: Δx (off-center gravity), r_v (void ratio), ρ_r (rupture/mark energy) and tells you if it sits within intended bands for its class. It's built to study **stability and failure** as you iterate, not to crown winners.

- **Balance** (Δx): How far the visual center is from the geometric center
- **Void** (r_v): The ratio of empty space to filled space
- **Detail** (ρ_r): The amount of edge energy and texture density in key areas

It combines these measurements into a 0-100 score and tells you whether an image "passes" basic compositional criteria structural exploration test.

→ This version now includes: **Color telemetry (advisory only)**. It logs a color-aware void mask and a luminance-only balance read to explain divergences. These **never gate** acceptance; they only trigger a one-line **Color audit** when the gap is material.

Why researchers use it (not FID/CLIP/SSIM).

Resemblance and caption metrics can't answer: *will this composition hold* when pushed? LSI-lite is a **composition structural exploration test** and a quick, profiled gate you can log across Baseline → Pressure → Collapse-trigger runs.

The Three Measurements

Δx (Delta-x): Off-Center Balance

- Finds the visual center of mass using edge detection
- Measures how far it is from the geometric center
- Range: 0.0 (perfectly centered) to 1.0 (at edge)
- **Why it matters:** Dead-center compositions often feel static

r_v (Void Ratio): Empty Space

- Calculates the percentage of background vs foreground
- Higher values = more empty space
- Range: 0.0 (completely filled) to 1.0 (mostly empty)
- **Why it matters:** Negative space creates visual breathing room
- → Color note: when color helps isolate subject/background, we also log a color-derived void read for comparison (see Color telemetry below).

ρ_r (Rho-r): Rupture Energy

- Measures texture and detail density around the main subject
- Uses Laplacian edge detection over a "halo" around the subject
- Range: 0.0 (smooth) to 1.0 (highly detailed)
- **Why it matters:** Controls visual complexity and focus

Research Context

The tool is designed to study compositional stability in AI-generated images. It focuses on structural principles under iteration/modification (not resemblance or caption alignment). The goal is not to replace human judgment, but to provide a systematic way to discuss why some compositions feel more stable than others outside of AI and human defaults. LSI-lite reports what shifts, not whether something is "good." If it still has structural stability, not if the center has no use reason.

Three Profiles

Different types of images need different compositional rules, this lite version covers:

Figure_Default

- **For:** Portraits, figure drawings, character art
- **Emphasis:** Balance (45%), then density (35%), then detail (20%)
- **Expects:** Moderate off-centering, balanced space usage, controlled detail

MarkMaking_Expressive

- **For:** Sketches, expressive artwork, gestural drawings

- **Emphasis:** More balanced weights (40%/30%/30%)
- **Expects:** Looser centering rules, higher texture tolerance

Landscape

- **For:** Landscape photos and paintings
- **Emphasis:** Density matters most (40%), then balance (35%), then detail (25%)
- **Special feature:** Detects water reflections and adjusts balance measurement accordingly
- → *Reflection check:* for strong water reflections, balance may be read on a shallower top window to avoid mirroring bias.

Gate rule (accept/reject).

- Combine the three measurements into LSI_lite_100 (0–100). **Accept** only if LSI_lite_100 ≥ 55 **and** no bands are **RED**; otherwise **reject** and name the limiting primitive (**priority_knob**).

Telemetry (advisory only)

Color audit thresholds (non-gating):

- Emit a Color audit only if the color-based void read differs **meaningfully** from gray, or the luminance-only balance read shifts **meaningfully** from gray.
- Rule of thumb: log an audit when the gap is **noticeable**; suppress when the reads agree. (*Engineering note for later: ≈ 0.15 void gap or ≈ 0.10 balance gap, or gray void sits right on a guard ± 0.02 .*)

How Scoring Works

1. **Measure:** Calculate all three primitives (Δx , r_v , p_r)
2. **Check bands:** Each primitive has an acceptable range - values outside are flagged "RED"
3. **Score:** Combine the three values using weighted geometric mean
4. **Gate:** Image "passes" if score ≥ 55 AND no RED flags
5. **(If warranted) Color audit.** If color-based reads diverge, add one line under the result. No change to accept/reject.

Example Output

What a single run gives you (at a glance).

- **Grade card:** profile • $\Delta x/r_v/p_r$ (0–1) • per-band OK/RED • LSI_lite_100 • **accepted** True/False • **priority_knob** (+ suggested raise/lower)
- **Stamp:** ACCEPT / REJECT / UNKNOWN — reason_for_fail=<enum>
- **Color audit** (*only when thresholds trip*):
- voids under color are lighter than gray; confirm subject mask / glare
- **Exports:** a composite PNG + one CSV/JSON row (fields below).

New fields (advisory):

- $r_v_color_mask$ — color-derived void read
- dx_color_L — luminance-only balance read
- Δr_v — difference between color void and gray void (*for engineers*)
- $\Delta \Delta x$ — difference between luminance balance and gray balance (*for engineers*)
- reason_for_fail — short enum: band_red, lsi_gate, unknown_input, calc_error, roi_mismatch, mask_empty
- color_audit_badge — **one line**; present only when an audit fires

What it is *not*.

Not aesthetic judgment. "Accepted" means "within intended structural bands," not "good art." Profiles reflect specific design intent; many excellent images will fail on purpose.

What This Tool Is Good For

- **Art education:** Consistent feedback on basic composition principles
- **Batch analysis:** Evaluate large sets of images quickly
- **AI research:** Study how different generation methods handle composition
- **Quality filtering:** Screen images for basic structural soundness

Important Limitations

Not Universal

- Built around Western compositional conventions
- Many excellent artworks will fail these tests intentionally
- Abstract art, experimental compositions may be penalized unfairly

Not Aesthetic Judgment

- "Accepted" means "follows basic rules," not "good art"
- Measures structure, not beauty, emotion, or meaning
- Doesn't evaluate color harmony/palette or meaning; **color telemetry is diagnostic only** (masking & luminance separation), not an aesthetic score.

This is not recognition or a "style police" or a judgement on "aesthetics." It's a tiny, defensible ruler over three compositional 101 primitives. Think of it as a quick ruler for balance, void, and stroke coherence. LSI-lite as a complementary metric in the generative AI evaluation ecosystem.

Current Scope

- Three primitives can't capture all compositional elements
- Profile bands are designer intuition, not empirically validated
- Works best on clear subject/background images

Technical Notes

- Handles EXIF rotation automatically
- Resizes to 1536px longest side for consistency
- Uses bilateral filtering and morphological operations for robustness
- Fallback handling for edge cases (no foreground detected, etc.)

"Limits"

Where Color Telemetry Can Mislead (watch-outs):

- Heavy cast lighting or LEDs can fake "void" where none exists.
- Glossy black or neon backgrounds may confuse the color mask.
- Busy fabrics/patterns around a centered subject can nudge the balance read.
(These are audit triggers, not automatic fails.)

Research Context

LSI-lite is designed as a research instrument to study compositional stability in AI-generated images. Unlike metrics that measure similarity to training data (FID) or text-image alignment (CLIP), this focuses on structural principles that observe how compositions hold up under iteration or modification. This also logs **color telemetry**, a color-aware void read and a luminance-only balance read, to explain divergences across materials and engines; these notes are **diagnostic only** and never affect acceptance.

The goal is not to replace human artistic judgment, but to provide a systematic way to discuss why some compositions feel more stable than others. LSI-lite reports **what shifts**, not why it's "good"; color telemetry is a compass note, not a verdict.

Artist's Statement

Structural Exploration. Artists don't march images through single images or a linear test; we explore a space of alternatives. Pressure is applied as a contradiction (e.g., torque↑ & permeability↑). The image may collapse, re-cohere, or discover a **new center**. LSI-lite doesn't score "beauty"; it logs whether off-center gravity (Δx), breathing ground (r_v), and mark energy (p_r) support the path you're taking. The guardrails aren't grades, they're **questions**: where does gravity sit, do voids breathe, are marks committed? Exploration ends when form is found or the gap is named.

How LSI is part of that story

- **Δx (gravity)** → "Did we escape centering in a way that reads as intentional gravity, not tip?"
 - *Color note (advisory):* when luminance isolates structure better than gray, we log a **luminance-only balance read** for comparison.
- **r_v (voids)** → "Are the alternatives ventilating, or sealing?"
 - *Color note (advisory):* when color separates subject/background, we also log a **color-derived void read**; it's reported only when it materially diverges from gray.
- **p_r (rupture)** → "Is energy committed, or decorative noise?"
- **Gate (≥ 55 , no RED)** → "Is this branch structurally sound *for this profile*?"

- *Telemetry does not gate.* Color reads may trigger a **one-line Color audit**; acceptance stays with the meter.
- **Priority knob** → “If we pursue this branch, what single nudge makes it truer to itself?”

The “aesthetic neutrality” is actually a feature, not a bug. By deliberately ignoring context, meaning, and style, it can flag violations of fundamental structural principles regardless of artistic intent. The system is doing its job by flagging deviation; the human artist/curator makes the call on whether that deviation is:

- Intentional choice (breaking rules)
- Cultural/stylistic context (asymmetry vs Western centering)
- AI default failure (model collapse into smoothed mediocrity)

Non-linear path (what we actually see)

- **Default → Collapse → Cohesion → New Center**
or
- **Default → Deviation → Poise (hold) → Deconstruction (on purpose)**
LSI marks the **state changes**, not a pass/fail morality.

What the AI does: Rapidly identifies structural patterns, deviations, mark-making quality.

What the human does: Distinguishes between intentional choice and algorithmic failure.

This is: Mathematical language for discussing artistic intention.

Color Telemetry — Watch-outs

- Heavy casts/LEDs can fake “void.” • Glossy blacks/neons can confuse the mask. • Busy patterns near center can nudge balance.
(These trigger an audit line only; they don’t fail the image.)

*It’s fieldwork in a landscape of alternatives. Each image is a branch; pressure asks whether the branch **finds form** or collapses into defaults (human and model). LSI isn’t judging taste; it’s the start of a **compass** that tells you where structure holds while you deviate.*

This is an experimental tool. Use it to explore patterns in images, not as a definitive judgment of artistic quality.

Appendix:

Methods (how it works, reproducible)

Primitives (what/why/how)

- **Δx — Centroid offset** (*off-center gravity / frame tension*)
How: bilateral denoise → **Canny(50,100)** → **edge centroid**; **fallback** to foreground centroid if edges empty. If both fail ⇒ **$dx = NaN$** (mark **unknown**, gate-fail).
- **r_v — Void ratio** (*figure-ground breath / negative space*)
How: Otsu threshold → largest component → open/close → **$rv = 1 - fill$** .
- **p_r — Rupture energy** (*mark/texture pressure*)
How: **Laplacian k=3** → abs → normalize → **mean over subject + halo** (halo fraction per profile).
- **Color telemetry (advisory only)**
How: derive a color-aware foreground mask (LAB k-means, k=3; union of non-background clusters).
Using that mask we log:
 - **rv_color_mask** — void ratio from the color mask
 - **dx_color_L** — Δx from the LAB L-channel (luminance)*Why:* explains divergences vs. gray reads; **never gates** acceptance.

Fields are logged even when the audit doesn’t fire; they’re for diagnosis only.

Profiles (guards & masks)

Each profile specifies weights and **band guards** (min..max). **Figure/MarkMaking** use a subject halo (0.08 / 0.12).

Landscape uses **rho_mask=full** and may apply **Δx ROI** via a reflection probe (top_frac=0.60) *for Δx only*. **r_v/p_r are always full-frame*.

Color telemetry is **profile-agnostic** (same thresholds for all profiles) and follows the same ROI decision used for Δx (e.g., Landscape reflection probe). It **does not** change band guards or weights.

Scoring

- Per-primitive **Gaussian in-band score** (0–1) from guard range.
- **K_lite** = weighted geometric mean of the three soft scores with $\epsilon=1e-6$ floor.
- **LSI_lite_100** = $100 \times K_lite$.
- **Accept** iff **LSI_lite_100 ≥ 55** and all bands **OK** and not **unknown**.
- Telemetry is **not** part of **K_lite** and does **not** affect **LSI_lite_100**. It only supports a one-line **Color audit** when warranted.

Quality-of-life diagnostics (for iteration)

Alongside $\Delta x/r_v/p_r$:

***_center** (band midpoint) • ***_to_center** • ***_to_center_norm** (0..1) • ***_direction** (“increase/decrease”) • **priority_knob** (which primitive is farthest from center) • **priority_score**.

Also emit: **mask_mode** (“gray” | “color” | “fallback_gray”), **color_space** (“LAB”), **color_status** (“ok” | “fail_mask” | “fail_convert” | ...) for quick triage.

Unknown semantics

If edges and foreground both fail: **dx = NaN**, **unknown=True**, bands = RED, gate-fail (this prevents silent “passes” on blank/edgeless frames).

Run loop (single pass)

Compute $\Delta x/r_v/p_r$ once, then compute telemetry (**rv_color_mask**, **dx_color_L**) and deltas:

delta_rv = **rv_color_mask** - **void_ratio**, **delta_dx** = **dx_color_L** - **delta_x**.

Build **color_audit_badge** only if thresholds trip; never recompute/overwrite Δx .

I/O and types

Masks are **uint8**; don't truth-test NumPy arrays (**if arr:**). Use **is None**, **np.any**, **np.all**. Runs local or Colab (no hard pins unless you need exact repro).

Color failures do **not** propagate to acceptance. If the color mask or L-read fails, set **color_status** accordingly and leave **rv_color_mask** / **dx_color_L** as **NaN**; no audit line is emitted (telemetry never gates).

Color telemetry follows the same Δx ROI decision (e.g., Landscape reflection probe) and never alters guards or weights.

What you get: Visualizations & Artifacts

Single-image visualization (3×3 grid)

1. Original (blue = geometric center; red = Δx centroid)
2. Grayscale used for analysis
3. Foreground mask (fill %)
4. Edges (Canny 50/100)
5. Halo mask preview (for p_r)
6. Laplacian energy heatmap (p_r field)
7. **Band gauges** ($\Delta x/r_v/p_r$): guard bar + center tick + measured marker + “increase/decrease”
8. **Gaussian curves** (tiny sparklines per primitive) with current value plotted
9. **Acceptance stamp** (**ACCEPT** / **REJECT** / **UNKNOWN**) + one-line reason for fail
Optional overlays: reflection ROI (Δx only), value histogram.
10. Color mask preview (over gray)

11. Luminance balance line (L-only Δx overlay)
12. **Color audit badge** (only if thresholds trip; one line under the stamp)

Exports:

- **PNG** of the grid
- CSV/JSON row per image with this schema (order flexible):
- `delta_x`, `void_ratio`, `rupture_rho`, `K_lite`, `LSI_lite_100`
- `band_delta_x`, `band_r_v`, `band_rho_r`, `accepted`, `frame`, `profile`
- `dx_inside_band`, `dx_to_center`, `dx_to_center_norm`, `dx_direction`
- `rv_inside_band`, `rv_to_center`, `rv_to_center_norm`, `rv_direction`
- `rho_inside_band`, `rho_to_center`, `rho_to_center_norm`, `rho_direction`
- `priority_knob`, `priority_score`
- **Color telemetry:** `rv_color_mask`, `dx_color_L`, `delta_rv`, `delta_dx`, `mask_mode`, `color_space`, `color_status`
- **UX:** `reason_for_fail` (enum: `band_red`|`lsi_gate`|`unknown_input`|`calc_error`|`roi_mismatch`|`mask_empty`), `color_audit_badge` (one line; present only when audit fires)
- `module` — "`engine/ridep`" or "`stabilizer/ridp`" when other modules write rows (leave empty for pure LSI runs).

Small watch-outs (drop near the export section, optional)

Color Telemetry Watch-outs: heavy color casts/LEDs can fake "void"; glossy blacks/neons can confuse masks; dense patterned fabrics near center can nudge the balance read. These trigger an **audit**, not a fail.

Batch visualization (dataset)

- **Score histogram** (+ gate line @55, acceptance rate)
- **Band failure breakdown** (% RED by $dx/r_v/\rho_r$)
- **Priority-knob distribution** (what limits most, and direction $\uparrow/\downarrow$)
- **Δx vs r_v scatter** (colored by LSI; target-zone overlay)
- **Correlation heatmap** (Δx , r_v , ρ_r , LSI)
- **Box/violin by profile** (primitive distributions across Figure / MarkMaking / Landscape)
- **Seed-stability view** (variance across seeds for the same prompt)
- **Failure atlas tiles** (top examples per Category-Ledger tag)
- **Systematic Gap table** (prompt $\rightarrow$ acceptance $\rightarrow$ common failure $\rightarrow$ gap note)
- **Color audit rate** (share of rows with an audit per profile)
- **Delta distributions** (`delta_rv`, `delta_dx` hist/violin per profile)
- **Telemetry failure reasons** (bar of `color_status` counts)

Use in studies (how to report, minimal)

- **Profile used** + **gate rule** (≥ 55 and no RED).
- **Acceptance rate**, **band-failure mix**, and **priority-knob distribution**.
- **Seed stability** for repeated prompts.
- **Systematic Gap** notes when language lacks a visual schema (optional but recommended).
- Include a **Color audit** line **only** when telemetry diverges; otherwise omit any mention of color.

Threshold note (footnote-style, don't headline it)

Rule-of-thumb trigger for the audit (engineering guidance):

`|delta_rv|  $\geq$  0.15` or `|delta_dx|  $\geq$  0.10`, or gray r_v sits within ± 0.02 of a guard. Keep these thresholds out of scoring; they're advisory.

One-liner (Methods box):

We use LSI-lite, a profiled composition structural exploration that scores off-center gravity (Δx), void ratio (r_v), and rupture energy (ρ_r) against profile band guards. A weighted geometric mean (ϵ -floored) yields `LSI_lite_100`; an image passes if `LSI_lite_100  $\geq$  55` and all bands are OK; edgeless/foregroundless frames are flagged unknown and fail. Profiles: `Figure_Default`, `MarkMaking_Expressive`, `Landscape`; a reflection probe may apply a Δx ROI in Landscape only.

Limits

- **Diagnostic, not aesthetic.** “Accepted” ≠ “good art.”
- **Profiled, not universal.** Western composition bias; many great images will fail **on purpose**.
- **Scope.** Color, concept, cultural context are out-of-scope; three primitives can’t capture all structure.

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Authorship

This framework was architected by Russell Parrish and recursively co-developed inside GPT, Gemini and Claude. Every critique is human-led; every recursion is model-driven. The result: a reasoning layer authored through language, not image manipulation.